# Integration by Parts

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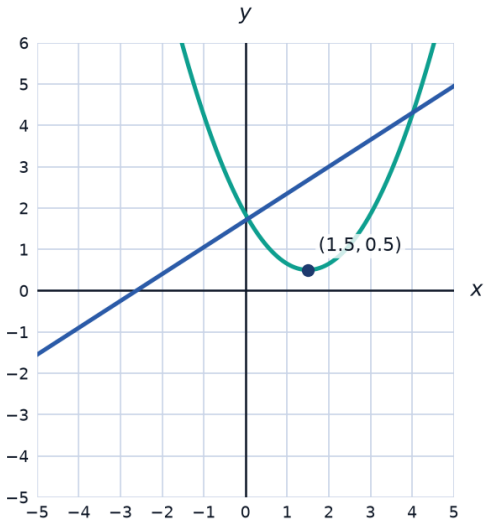
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: The integration by parts formula

Evaluate $\int xe^x\,dx$ using integration by parts.

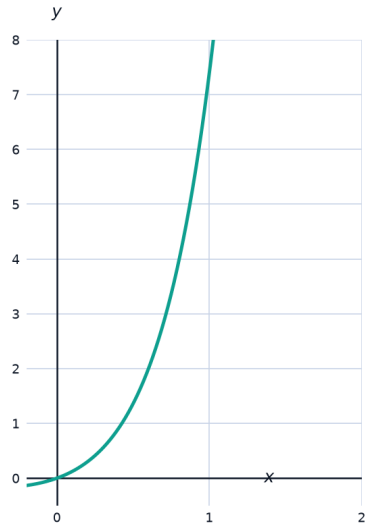
**2** [Foundational] • Target: Repeated and definite applications

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.




**3** [Foundational] • Target: Repeated and definite applications

Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.




**4** [Foundational] • Target: Choosing u and dv wisely (LIATE)

Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as $u$ under the LIATE rule).

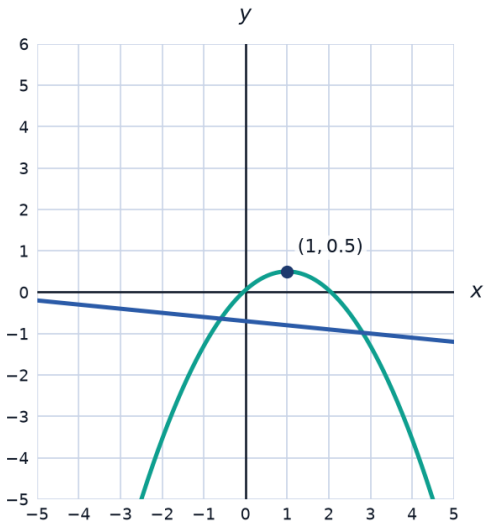
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**5** [Foundational] • Target: The recurring-integral trick

Evaluate $\int e^x \sin x \, dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

**6** [Foundational] • Target: Reduction formulas

Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x \, dx = x^n e^x - n\int x^{n-1} e^x \, dx$.




**7** [Foundational] • Target: Reduction formulas

Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x \, dx$, confirming it matches an earlier result.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Integration by parts with inverse trig functions

Evaluate $\int x \sin^{-1} x \, dx$ using $u = \sin^{-1} x$, $dv = x \, dx$ (the resulting integral can be left in terms of a known antiderivative form).

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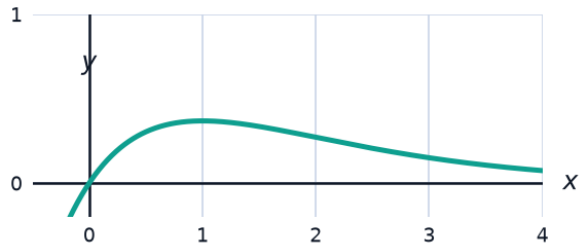
**9** **[Intermediate]** • **Target: A capstone mixed-technique problem**

Evaluate $\int_0^{\pi/2} x\cos(2x)\,dx$ exactly, using integration by parts.




**10** **[Intermediate]** • **Target: Visualizing an area found by integration by parts**

The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.




**11** **[Intermediate]** • **Target: Choosing between substitution and parts**

For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice ($u$-substitution variable, or $u/dv$ for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

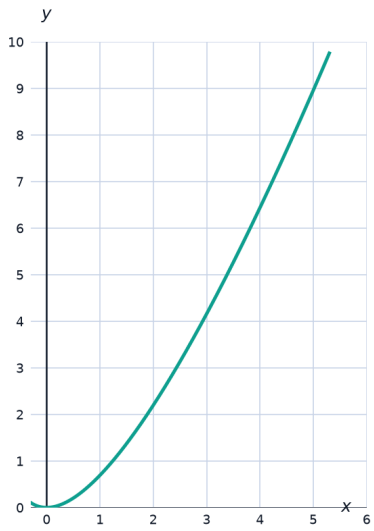
**12** **[Intermediate]** • **Target: Choosing between substitution and parts**

Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

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**13** [Intermediate] • Target: Choosing between substitution and parts

A city's population growth rate is modeled by $r(t) = t\ln(t+1)$ thousand people per year, for $0 \le t \le 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.




**14** [Intermediate] • Target: The recurring-integral trick

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \ge 0$. (a) Evaluate $I_0$. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \ge 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

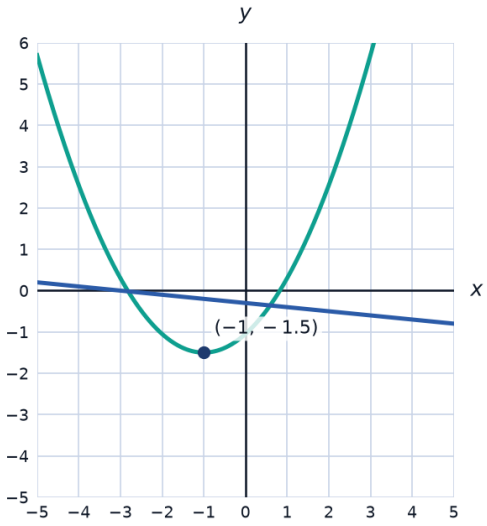
**15** [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 2.02 Continuity

Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

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**16** **[Challenging] • Target: Reduction formulas • Cross-Lesson: 3.01 The Derivative from First Principles**

[Unit 8 · Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.




**17** **[Challenging] • Target: Integration by parts with inverse trig functions • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^{\infty} xe^{-2x}\,dx$ or establish that it diverges.

**18** **[Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

Level 2 · MediumFind the equations of all horizontal and vertical tangent lines to the ellipse $4x^2 + 9y^2 = 36$.

**19** **[Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Level 2 · MediumThe base of a solid is bounded by $y = \sqrt{x}$, the line $x = 9$, and the x-axis. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume of the solid.

**20** **[Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

Level 1 · FoundationsEvaluate the indefinite integral $\int \arctan(x)\,dx$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: The integration by parts formula

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

**a.** $\frac{5\pi}{16}$

**b.** $\frac{5\pi}{32}$

**c.** $\frac{3\pi}{16}$

**d.** $\frac{1}{6}$

**22** [Foundational] • Target: Repeated and definite applications

Reduction formula: $I_n = \int \sin^n x\,dx$ satisfies:

**a.** $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$

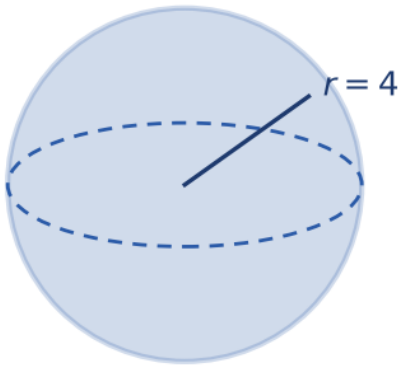
**b.** $I_n = \frac{\sin^{n+1} x}{n+1} + C$

**c.** $I_n = -n\cos x \sin^{n-1} x + C$

**d.** $I_n = I_{n-1} \cdot I_1$

**23** [Foundational] • Target: Choosing u and dv wisely (LIATE)

Volume between paraboloids $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$:




**a.** $4\pi$

**b.** $8\pi$

**c.** $16\pi$

**d.** $32\pi/3$

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**24** [Foundational] • Target: The recurring-integral trick

Consumer surplus formula (demand $p = D(q)$, equilibrium $(q^*, p^*)$):

**a.** $\int_0^{q^*} [D(q) - p^*]\,dq$

**b.** $\int_0^{p^*} D(q)\,dq$

**c.** $p^* q^* - \int_0^{q^*} D(q)\,dq$

**d.** $D(q^*) - p^*$

**25** [Foundational] • Target: Definite integrals by parts

Reduction formula for $\int \sin^n x\,dx$ comes from IBP with $u =$

**a.** $\sin^n x$

**b.** $\cos x$

**c.** $\sin^{n-1} x$

**d.** $x$

**26** [Foundational] • Target: Reduction formulas

Tabular integration (repeated parts): $\int x^3 e^x\,dx$. Number of IBP steps needed:

**a.** 2

**b.** 1

**c.** 4

**d.** 3

**27** [Foundational] • Target: Integration by parts with inverse trig functions

Archimedes' principle as an integral: buoyancy force = weight of fluid displaced = $\rho g$ times:

**a.** surface area

**b.** volume of submerged body

**c.** pressure × depth

**d.** mass of body

**28** [Foundational] • Target: A capstone mixed-technique problem

Surface area of revolution about x-axis: $S =$

**a.** $\pi \int_a^b f(x)\sqrt{1 + f'(x)}\,dx$

**b.** $2\pi \int_a^b f(x)\sqrt{1 + [f'(x)]^2}\,dx$

**c.** $2\pi \int_a^b [f'(x)]^2\,dx$

**d.** $\int_a^b f(x)dx$

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Visualizing an area found by integration by parts**

Area of surface of revolution $r = f(\theta)$ about polar axis:

**a.** $2\pi\int f\sin\theta\sqrt{f^2 + f'^2}\,d\theta$

**b.** $\pi\int f^2\,d\theta$

**c.** $2\pi\int f\,d\theta$

**d.** $4\pi\int f\cos\theta\,d\theta$

**30** **[Intermediate]** • **Target: Choosing between substitution and parts**

Gauss-Green formula in 2D: $\iint_\Omega abla\cdot\mathbf{F}\,dA =$

**a.** $\oint_{\partial\Omega}\mathbf{F}\cdot\mathbf{n}\,ds$

**b.** $\oint_{\partial\Omega}\mathbf{F}\cdot d\mathbf{s}$

**c.** $\iint_\Omega|\mathbf{F}|\,dA$

**d.** $0$

**31** **[Intermediate]** • **Target: The integration by parts formula**

$\int\sin^4 x\,dx$ using reduction formula:

**a.** $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

**b.** $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

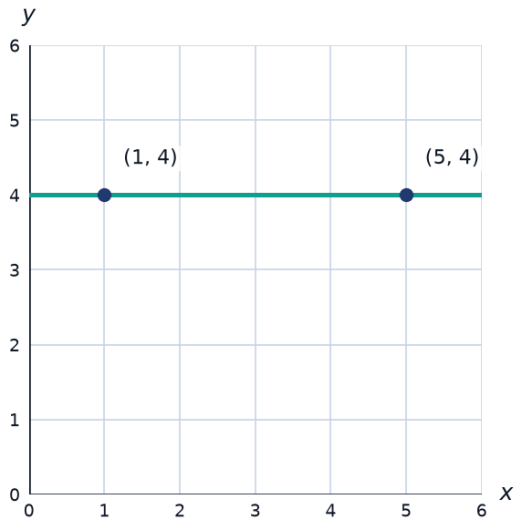
**c.** $\frac{x}{2} - \frac{\sin 4x}{8} + C$

**d.** $\frac{\sin^3 x\cos x}{4} + \frac{\sin x\cos x}{4} + \frac{x}{8} + C$

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**32** [Intermediate] • Target: Repeated and definite applications

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.




**a.** 17

**b.** 21

**c.** 9

**d.** 11

**33** [Intermediate] • Target: Choosing u and dv wisely (LIATE)

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

**a.** $\frac{1}{x-1} + \frac{1}{(x-1)^2}$

**b.** $\frac{x}{(x-1)^2}$

**c.** $\frac{1}{x-1}$

**d.** $\frac{2}{(x-1)^2}$

**34** [Intermediate] • Target: The recurring-integral trick

Arc length in parametric form ($x = x(t), y = y(t)$):

**a.** $\int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$

**b.** $\int_a^b \sqrt{(dx/dt) + (dy/dt)}\, dt$

**c.** $\int_a^b |dy/dx|\, dt$

**d.** $\sqrt{x(b)^2 + y(b)^2}$

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**35** [Intermediate] • Target: Definite integrals by parts

Kinetic energy of rotating rod (length $L$, mass $M$, angular velocity $\omega$):

**a.** $\frac{1}{3}M\omega^2L^2$

**b.** $\frac{1}{6}M\omega^2L^2$

**c.** $\frac{1}{2}M\omega^2L^2$

**d.** $M\omega^2L^2$

**36** [Intermediate] • Target: Reduction formulas

Force on a flat plate submerged at depth $d$, width $w(y)$, density $\rho$:

**a.** $\rho g\int w(y)(d-y)\,dy$

**b.** $\rho g dw$

**c.** $\rho w d^2$

**d.** $\int w\,dy$

**37** [Intermediate] • Target: Integration by parts with inverse trig functions

Moments: $M_y = \rho\int_a^b x[f(x)-g(x)]dx$. Centroid $\bar{x} = M_y/M$ where $M =$

**a.** $\rho\int_a^b[f(x)-g(x)]dx$

**b.** $\rho(b-a)$

**c.** $\rho \cdot A$ (same as first)

**d.** $M_y$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Probability of event $a \le X \le b$ for PDF $f$:

**a.** $\int_a^b f(x)\,dx$

**b.** $f(b)-f(a)$

**c.** $f(a)\cdot(b-a)$

**d.** $F(b)\cdot F(a)$

**39** [Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Centroid of semicircular region radius $R$, $y \ge 0$:

**a.** $\bar{y} = R/2$

**b.** $\bar{y} = 4R/(3\pi)$

**c.** $\bar{y} = 2R/3$

**d.** $\bar{y} = R/\pi$

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**40** [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

A cylindrical tank of radius 3 m and height 2 m is full of water. Write the work to pump all the water to a point 1 m above the top as $W = k\rho g$. Find $k$.

**a.** $36\pi$

**b.** $54\pi$

**c.** $36$

**d.** $18\pi$

**41** [Challenging] • Target: The integration by parts formula • Cross-Lesson: 6.01 Related Rates

Work done lifting a 10 kg mass 4 m ($g = 10$ m/s²):

**a.** 400 J

**b.** 40 J

**c.** 4 J

**d.** 100 J

**42** [Challenging] • Target: Repeated and definite applications • Cross-Lesson: 6.02 Optimization and Curve Sketching

Volume of revolution of $y = x^3$ about x-axis from 0 to 1:

**a.** $\pi$

**b.** $\pi/5$

**c.** $\pi/7$

**d.** $3\pi/7$

**43** [Challenging] • Target: Choosing u and dv wisely (LIATE) • Cross-Lesson: 7.01 Integration by Substitution

Moment of inertia of disk radius $R$, mass $M$ about its axis:

**a.** $MR^2$

**b.** $\frac{1}{4}MR^2$

**c.** $\frac{1}{2}MR^2$

**d.** $2MR^2$

**44** [Challenging] • Target: The recurring-integral trick • Cross-Lesson: 8.01 Area Between Curves

Find the angle between $\mathbf{u} = \langle 1, 2, -4 \rangle$ and $\mathbf{v} = \langle 4, -2, 0 \rangle$.

**a.** $\frac{\pi}{4}$

**b.** $\frac{\pi}{3}$

**c.** $\frac{\pi}{6}$

**d.** $\frac{\pi}{2}$

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**45** [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 8.02 Volumes of Revolution

Spring: force $F = kx$. Work stretching from 0 to $d$:

**a.** $kd^2/2$

**b.** $kd$

**c.** $kd^2$

**d.** $2kd$

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